

AMITY UNIVERSITY

-----UTTAR PRADESH-----

Amity School of Engineering and Technology

Minor Project

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Industry Guide

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Project Information

1) Project Duration : (82 Days)

- a) Date of Summer Internship commencement (20/07/2026)
- a) Date of Summer Internship Completion (09/10/2026)

2) Topic

Visual Microphone Pro: Sub-Pixel Acoustic Reconstruction from Video Streams using GPU-Accelerated Quaternionic Riesz Pyramids and Neural Speech Enhancement

3) Project Objective

To recover clear, intelligible audio from sub-pixel surface vibrations in video streams by developing a GPU-accelerated Quaternionic Riesz Pyramid pipeline enhanced with motion stabilization, rolling shutter super-resolution, and deep learning speech denoising.

4) Methodology to be adopted

Pre-Processing & Video Registration: Frames are stabilized using Enhanced Correlation Coefficient (ECC) alignment to decouple macro camera movements from microscopic vibrations. Spatial-Temporal Phase Decomposition: Each frame is decomposed into a multi-scale Laplacian pyramid. A 2D Riesz transform (central difference operators) extracts local quaternionic phase and orientation. Temporal Phase Filtering & Unwrapping: Local frame-to-frame phase differences are unwrapped and temporally filtered through an IIR bandpass filter tuned to target speech/acoustic frequencies. Rolling Shutter Super-Resolution: For consumer cameras, row-by-row CMOS readout timing is exploited to temporally upsample pixel rows into microsecond acoustic samples ($F_{\text{readout}} = \frac{\text{Frame Height}}{\text{FPS}}$). Phase Fusion & AI Post-Processing: Multi-scale phase signals are fused using local SNR weighting. The composite 1D signal is passed through a deep speech enhancement neural network to output clean, intelligible audio (.wav).

5) Brief Summery of project(*to be duly certified by the industry guide*)

This project presents an advanced Python-based computer vision system that recovers audible acoustic signals from high-speed and standard consumer video sequences without physical microphones. Sound waves hitting physical objects (such as chip bags, plant leaves, or glass) cause high-frequency, sub-pixel surface vibrations. By extending MIT's original Visual Microphone concept, this project develops a GPU-accelerated pipeline utilizing Quaternionic Riesz Pyramids for phase-based temporal motion analysis. To overcome the fundamental limitations of prior hand-crafted MATLAB implementations—namely high computational latency, vulnerability to camera shake, strict high-FPS requirements, and sensor noise—this project introduces: 1. Camera Motion Decoupling via Enhanced Correlation Coefficient (ECC) image registration. 2. CMOS Rolling Shutter Temporal Super-Resolution to recover acoustic frequencies exceeding the Nyquist limit of consumer 30/60 FPS cameras. 3. Deep Learning-Based Speech Enhancement to eliminate camera read noise and environmental interference. 4. An End-to-End Production-Grade Interactive Web/Desktop Application providing real-time audio extraction, region-of-interest (ROI) tracking, and STFT spectral visualization.

Signature
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